



TECHNOLOGY AND LIVELIHOOD EDUCATION

IA - AUTOMOTIVE AND SMALL ENGINE SERVICING

Grade 9 or 10

**Three-Term Budget Of Work (BOW)
for Learning Competencies**

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IA
Grade 9 and Grade 10

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For Learning Competencies

Please note that each sector under TLE: IA is studied for one term. This is applicable under Grades 9 and 10.

Term		
Sector	Automotive and Small Engine Servicing	
Performance Standard	The learners perform ignition system servicing following the safety precautions. The learners perform engine servicing with safety precautions. The learners compute costs for labor and materials for services.	
Content Standard	Demonstrate an understanding of the concepts and principles of ignition system servicing in automotive and small engines.	
Week	Learning Competency	Suggested Activities
1	Interpret the fundamentals in automotive and small engine services.	Engine System Identification and Function Mapping - Learners examine actual engine parts, models, or diagrams and match each component to its correct engine system and function. - Observe the engine parts or diagrams and explain which system each part belongs to and what role it plays in engine operation.

		• Basic Engine Trouble Interpretation Scenario ○ Learners analyze simple automotive or small engine problem scenarios and determine which engine system is responsible based on symptoms. ○ Read the engine problem scenario and interpret which system is involved and why, using basic engine principles.
2	• Discuss tools, materials, consumables, and equipment.	• Tool, Material, and Equipment Classification Activity ○ Learners examine real shop items or images and classify them as tools, materials, consumables, or equipment, explaining their basic use. ○ Group the given shop items into tools, materials, consumables, or equipment and briefly explain the purpose of each. • “What Do I Use?” Workshop Planning Task ○ Learners plan a simple Industrial Arts task (e.g., oil change, engine tune-up, or parts cleaning) by identifying the correct tools, materials, consumables, and equipment needed. ○ List and discuss the appropriate tools, materials, consumables, and equipment required to complete the given workshop task.

3	Distinguish the component parts of automotive and small engine systems.	Electrical Components Identification and Labeling - Learners examine actual electrical parts or diagrams (battery, ignition coil, spark plug, wiring, switches, fuses) and label each component according to its function. - Identify and label each electrical component shown and state its basic function in the engine's electrical system. Simple Electrical System Tracing Activity - Learners trace the flow of electricity in a basic automotive or small engine electrical diagram from the power source to the load. - Follow and explain the path of electrical current in the given engine diagram, naming each component it passes through.
4	Draw a diagram of an engine electrical system.	Guided Engine Electrical System Diagram Drawing - Learners study a sample engine electrical system (battery-switch-ignition-spark plug) and redraw it using correct symbols and labels. - Copy and draw the engine electrical system diagram using proper symbols and label each component correctly.

		• Build-and-Draw Electrical System Activity ○ Learners assemble a simple engine electrical circuit using pictures or actual components, then sketch the system they created. ○ After arranging the electrical components, draw a diagram that shows the correct connection and flow of electricity.
5	• Describe the parts and functions of the ignition system.	• Ignition System Parts Identification and Function Chart ○ Learners examine real ignition system parts or diagrams (battery, ignition switch, coil, distributor/CDI, spark plug) and complete a parts-and-functions chart. ○ Identify each ignition system part shown and describe its specific function in producing spark for the engine. • Ignition System Sequence Explanation Activity ○ Learners arrange ignition system components in the correct operating order and explain how electricity flows to create ignition. ○ Arrange the ignition system parts in order and explain how each part works together to ignite the air-fuel mixture.

6	• Apply servicing of the ignition system following the safety precautions.	• Safe Ignition System Inspection and Cleaning ◦ Learners perform a guided inspection and basic cleaning of ignition system components (spark plug, wiring, ignition coil) while observing proper safety practices. ◦ Inspect and clean the ignition system components by following the correct servicing steps and safety precautions. • Spark Plug Removal, Checking, and Reinstallation ◦ Learners safely remove, inspect, gap, and reinstall a spark plug using the proper tools and personal protective equipment (PPE). ◦ Remove, check, and reinstall the spark plug correctly while strictly observing safety procedures.
7	• Differentiate types of engine operation.	• Engine Operation Comparison Chart Activity ◦ Learners compare different engine operations (2-stroke vs 4-stroke, gasoline vs diesel) using a guided comparison chart. ◦ Complete the comparison chart by identifying the differences in engine operation based on strokes, fuel use, and power cycle. • Stroke Cycle Sequencing and

		Explanation ○ Learners arrange illustrated cards showing engine strokes in the correct order and explain how each stroke operates. ○ Sequence the engine operation images correctly and explain what happens during each stroke.
8	• Describe engine construction.	• Engine Parts Identification and Assembly Mapping ○ Learners study engine diagrams or actual engine parts and identify major components (cylinder block, piston, crankshaft, valves, head) and how they fit together. ○ Identify the major engine parts and describe how they are assembled to form a complete engine. • Cutaway Engine Observation and Explanation ○ Learners observe a cutaway engine model or video and explain how the internal parts are arranged inside the engine. ○ Observe the cutaway engine and describe the construction and placement of its internal components.
9	• Apply engine servicing following the safety precautions.	• Safe Engine Inspection and Basic Servicing Activity

		○ Learners perform a guided engine inspection and basic servicing tasks (checking oil level, tightening bolts, cleaning parts) while observing proper safety procedures. ○ Carry out basic engine servicing tasks by strictly following the correct steps and safety precautions. ● Engine Servicing Safety Checklist Demonstration ○ Learners demonstrate proper use of tools, PPE, and safe work practices while servicing an engine using a safety checklist. ○ Service the engine by following the safety checklist and demonstrating correct tool use and personal protection.
10	● Calculate labor and material costs for services.	● Workshop Service Cost Computation Activity ○ Learners are given a sample service job (e.g., oil change or small engine tune-up) with labor rates and material prices and calculate the total cost. ○ Compute the total cost of a service by adding labor charges and material expenses using the given rates. ● Service Job Quotation Simulation ○ Learners create a mock quotation for a service job by determining labor

		hours, labor rate, and materials needed with their prices. ○ Prepare a service quotation by calculating labor costs and material costs accurately for the assigned task.
Suggested Performance Tasks	● Individual Performance Task - Safe Ignition System Servicing ○ The learner independently inspects and services an ignition system by identifying components, checking for faults, and performing basic servicing while strictly observing safety precautions. ○ Individually inspect and service the ignition system by following proper procedures and safety precautions. ● Group Performance Task - Team-Based Ignition System Servicing ○ In small groups, learners work together to inspect, service, and demonstrate proper ignition system servicing procedures, emphasizing teamwork and safety compliance. ○ As a group, perform ignition system servicing and demonstrate correct procedures while observing all safety precautions. ● Individual Performance Task : Engine Servicing with Safety Compliance ○ The learner independently performs basic engine servicing such as checking oil level, inspecting filters, and tightening loose components while strictly observing safety precautions and proper tool use. ○ Individually perform basic engine servicing by following correct procedures and observing all safety precautions. ● Group Performance Task : Engine Servicing and Safety Demonstration ○ In groups, learners work together to carry out engine servicing tasks and demonstrate proper safety practices, including correct tool handling, PPE use, and hazard prevention. ○ As a group, service the engine and demonstrate proper procedures while applying all required safety precautions. ● Individual Performance Task - Service Cost Computation	

	○ The learner independently computes the total cost of a given service by calculating labor charges and material costs based on provided rates, quantities, and service requirements. ○ Individually compute the total service cost by calculating labor and material expenses using the given data. ● Group Performance Task : Service Cost Estimation and Breakdown ○ In groups, learners analyze a service scenario and collaboratively compute labor and material costs, then present a clear and organized cost breakdown for the service. ○ As a group, calculate and present the labor and material cost breakdown for the assigned service scenario.
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